The Impact of Childcare Costs on Mothers' Labor Force Participation

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Abstract

The rising costs of childcare pose challenges for families, leading to difficult choices including those impacting mothers' labor force participation. This paper investigates the relationship between childcare costs and maternal employment. Using data from the National Database of Childcare Prices, the American Community Survey, and the Longitudinal Employer Household Dynamics, we estimate the impact of childcare costs on mothers' labor force participation through two empirical strategies. A fixed-effects approach controls for geographic and temporal heterogeneity in costs as well as mothers' idiosyncratic preferences for work and childcare, while an instrumental variables approach addresses the endogeneity of mothers' preferences for work and childcare by leveraging exogenous geographic and temporal variation in childcare licensing requirements. Our findings across both research designs indicate that higher childcare costs reduce labor force participation among mothers, with lower-income mothers exhibiting greater responsiveness to changes in childcare costs.

Keyword: childcare, female labor force participation, child penalty

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1 Introduction

Single-parent and dual-working parent households rely on affordable, quality childcare to maintain their productivity in the workforce. Yet, for American families with children not yet old enough to attend public school, childcare represents a significant financial burden, consuming up to 40% or more of household income (CCAoA, 2019; Birken et al., 2024). This financial strain may force parents to make difficult labor force participation decisions, such as reducing work hours or exiting the workforce entirely. Such adjustments, influenced by social norms and historical wage disparities, tend to disproportionately impact mothers (Ruppanner et al., 2021).

In contrast to many developed nations, the U.S. does not offer universally subsidized childcare. Instead, federal and state initiatives such as the Childcare and Development Block Grant Subsidy (CCDBG), Head Start, and tax credits aim to lessen the financial load of childcare. Subsidies are targeted at low-income families and are capacity-constrained, so they only benefit an estimated 14% of the eligible population (Chien, 2020; Chaudry et al., 2018). Tax credits like the Child and Dependent Care Tax Credit (CDCTC) are non-refundable, meaning that they help reduce tax liabilities to zero but any excess may not be received as a refund, and are only available for those who have childcare expenses and no non-working parent in the household; consequently, they only benefit around 13% of families with children.[1] Consequently, many low and middle-income families still face high childcare costs and must choose between lower-quality childcare and significant career setbacks.

In this study, we estimate the causal impact of childcare costs on the labor force participation of mothers. Studying this relationship presents empirical challenges, from research design to data availability. While we hypothesize that the cost of non-parental care significantly influences a mother's choice to work, this relationship is subject to reverse causality. As more women enter the labor force, the demand for childcare rises, driving childcare prices

[1] See Urban Institute and Brookings Institution Tax Policy Center brief: `https://www.taxpolicycenter.org/briefing-book/how-does-tax-system-subsidize-child-care-expenses`. Accessed July 29, 2024.

up. Additionally, other factors may simultaneously affect both the cost of childcare and a mother's decision to join the labor force such as the supply of childcare, subsidy reimbursement rates, work requirements for childcare assistance, and increases in the minimum wage. To address these identification issues, we employ two empirical strategies using a rich dataset that we construct by linking U.S. Census restricted-access data from the American Community Survey (ACS) with the Longitudinal Employer-Household Dynamics (LEHD), childcare regulations data from licensing studies conducted by the Inter-University Consortium for Political and Social Research (ICPSR), and the National Database of Childcare Prices (NDCP). The NDCP, published by the U.S. Department of Labor - Women's Bureau, is the most comprehensive federal source of childcare market prices available at the county level.

Our empirical strategy addresses the aforementioned identification challenges in two important ways. First, we mitigate bias arising from unobserved geographic and temporal factors as well as mothers' idiosyncratic preferences for work and children by exploiting the panel structure of our data to estimate a fixed-effects model that allows us to control for spatial, temporal, and mothers' heterogeneity.[2] Second, we address reverse causality by employing an instrumental variables (IV) approach that estimates the impact of childcare costs on mothers' labor force participation, relying solely on exogenous variation in childcare costs due to changes in childcare regulations. We use three distinct childcare regulations—child-to-staff ratios, group size, and staff education and training requirements—as exogenous shifters of cost. Our findings indicate that mothers with children under age five are 1.4 to 0.6 percentage points less likely to be employed when facing higher childcare costs. Consistent with prior research, we also find that low-income mothers are more sensitive to changes in childcare prices than their higher-income counterparts.

Research from the 1990s and early 2000s forms the core of our understanding of childcare costs' impact on maternal labor force participation. A recent literature review of these

[2]Non-mover mothers allow us to also mitigate the bias of unobserved time-invariant geographic factors.

earlier studies shows that the predominant finding is that childcare costs deter mothers from entering the workforce, with elasticities ranging from -0.025 to -1.1, clustering around -0.05 to -0.25 (Morrissey, 2016). This variation likely stems from differences in data sources, sample compositions, time periods, and methodologies, highlighting the complex and dynamic nature of the relationship between childcare costs and maternal employment.

During this earlier period of research, scholars relied mostly on survey data of childcare expenditures due to the lack of childcare market price data. Cross-sectional research shows a negative and significant effect of childcare expenses on mothers' labor force participation, with a greater response from single and lower-income mothers (Tekin, 2007; Han and Waldfogel, 2001). Longitudinal studies expand our understanding by showing that the negative impact of childcare costs diminishes over time as children age, suggesting a dynamic relationship between childcare costs and labor supply (Baum, 2002).

A second body of research emerged in the early to mid-2000s focusing on the impact of implicit or direct childcare subsidies on maternal labor force participation. Studies using public schooling as an implicit subsidy find mixed evidence, with earlier research showing a positive impact on mothers' labor force participation (Gelbach, 2002; Baker et al., 2008), while more recent studies find weaker or no impacts (Fitzpatrick, 2012; Havnes and Mogstad, 2011; Cascio, 2009; Cascio and Schanzenbach, 2013). Some key factors contributing to differences among these studies include the specific sample of mothers considered (domestic versus international), differences in time periods, as well as differences in empirical design. Similarly, studies analyzing the impact of direct childcare subsidies, such as Head Start and the Childcare and Development Block Grant (CCDBG), face empirical challenges that result in mixed evidence either due to the additional services provided or the employment requirements tied to the subsidies (Blau and Tekin, 2007; Pihl, 2022; Gurrentz, 2022).

Recent literature has shed light on the complex relationship between childcare costs and maternal labor force participation in the contemporary labor market. Ruppanner et al. (2019) and Landivar et al. (2022) highlight the barriers that expensive childcare poses to

maternal labor supply, though they note that their findings can not be interpreted as causal. Causal estimates from current research exploit variation in tax credit programs, supporting previous findings of a positive effect on mothers' labor force participation due to reductions in childcare costs (Neumark and Shirley, 2020; Pepin, 2020; Erceg, 2023).

Our study makes significant contributions to this evolving body of literature, with a particular focus on childcare costs for young children and the impact of those costs on maternal labor force participation. By leveraging our access to rich demographic and employment information from the ACS-LEHD data, and comprehensive information on childcare prices, we provide a direct and policy-relevant analysis that addresses some of the limitations in existing research.

A key contribution of our study lies in its narrow focus on early childcare costs, specifically for children below age five. This targeted approach allows us to more precisely capture the effect of childcare expenses on the labor force participation of mothers with young children. Unlike studies that rely on tax credit expansions applicable to parents of children up to age 13, our identification strategies leverage direct variation in childcare costs for the youngest age group. This precision enables us to isolate the impact of costs specifically related to children below school age, offering a clearer understanding of how these expenses influence mothers' labor supply decisions during a critical period of child development.

Furthermore, our use of the most comprehensive dataset on childcare market prices allows us to study the impact of childcare prices on maternal labor force participation for a broader set of mothers, without restricting our analysis to specific income groups. This approach provides a more comprehensive understanding of the role of childcare in shaping mothers' labor outcomes and the overall economy.

This paper proceeds as follows: Section (2) describes the different sources and data used for analysis and the linking process. Section (3) presents the empirical framework. Section (4) presents the results and discussion. Section (5) concludes.

2 Data

Our study combines data from several national surveys and administrative databases to create a comprehensive dataset. We draw information from the restricted-access American Community Survey (ACS) and the Longitudinal Employer-Household Dynamics (LEHD) program, the National Database of Childcare Prices (NDCP), the Childcare Licensing Studies series, Head Start Program Information Reports (PIR), U.S. Census County Business Patterns (CBP), and the Non-employer Statistics by Demographics (NES-D). This approach allows us to examine various factors related to childcare costs and maternal employment across different geographic and demographic contexts. In the following sections, we describe each data source in detail, outlining the specific variables and information extracted for each of our analyses.

2.1 ACS-LEHD

The American Community Survey (ACS) is a nationally representative survey that collects data on the social, economic, demographic, and housing characteristics of the U.S. population on an annual basis. We use confidential ACS 1-year data to create a dataset of mothers and the number and ages of their children. We also obtain ACS information on other control variables including the mother's age, educational attainment, marital status, race, and ethnicity. In line with the period for which childcare licensing data are available, we use ACS 1-year data for the years 2012 to 2017. Some mothers appear multiple times in the data due to the possibility of being resurveyed in the ACS. In such cases, we retain the most current record and drop all others.

We leverage additional confidential Census data from the Longitudinal Employer-Household Dynamics (LEHD) data for years 2012 to 2017, for information on mothers' earnings by quarter. The LEHD is an employer-employee matched dataset compiled by the Census Bureau from earnings data from the state Unemployment Insurance (UI) offices, the Quarterly

Census of Employment and Wages (QCEW), and previously collected survey data on jobs, businesses, and workers to create a national, quarterly data series that covers 95% of the private-sector workforce in the United States (Abowd et al., 2009).

We measure earnings in these data as total earnings in a quarter, combining earnings records from multiple employers, when applicable. With this information, we define labor force participation as any positive earnings in a given quarter. Note that, due to the fact that hours of labor are not included in the LEHD data, our analysis can only examine the impacts of the costs of childcare on the extensive margin of labor force participation. The intensive margin of mothers' labor force participation, along which childcare adjustments are also made, is not observable in this study. In addition, we cannot observe in the LEHD whether one is actively seeking work. Therefore, our labor force participation indicator differs from the canonical definition of labor force participation in that it excludes those who are technically participating in the labor force through their search for employment.

We link the ACS data to the LEHD via a Census-assigned Personal Identification Key (PIK), a unique number that allows us to link people across datasets while protecting their identities (Wagner and Lane, 2014). The resulting combined ACS-LEHD data contains information on mothers, their quarterly earnings, the quarterly earnings of their spouses, and the age(s) and number of their children.

To ensure complete information on household children and their ages, we retain the ACS-LEHD matched quarter along with all its preceding quarters from the LEHD panel. This results in an unbalanced panel, where we observe all mothers starting in 2012, but each mother exits the panel at different times based on her ACS survey date. Using ACS data on the number of children and their birth dates, we construct a panel tracking each mother's children and their corresponding ages backwards in time. We then restrict the sample to mothers who have at least one child under age five at some point during the study period. As a result, some mothers in our sample are observed in quarters prior to having children, and some mothers are observed after their child(ren) age(s) out of the cost-intensive childcare

years; however, no mother is observed beyond her ACS survey date.

There are some control variables for which we can only obtain cross-sectional data, including the mothers' education, race and ethnicity, and marital status. To exploit variation from these additional cross-sectional variables, we conduct additional analyses in the cross-section of the data that is generated by linking the ACS to LEHD and retaining data only for the quarter of the ACS-LEHD match. We additionally restrict this sample to mothers who have a single child under five years of age in the ACS-LEHD matched quarter.

2.2 National Database of Childcare Prices

The National Database of Childcare Prices (NDCP) is a public dataset of market childcare prices published by the U.S. Department of Labor Women's Bureau. The NDCP data originates from state childcare market rate surveys (MRS), which collect price data on all regulated full-time childcare providers, including center-based and family-based care providers. As part of their participation in the Child Care and Development Fund program, states must complete these MRS reports every three years. Although the MRS are mandatory, there is no standard format or methodology for collecting information. Consequently, the Department of Labor Women's Bureau harmonizes the collected price data from most states in the NDCP. The exceptions are Indiana and New Mexico, which have formats and methodologies that vary too much from the rest. This results in missing childcare price data for Indiana and New Mexico in the NDCP (Department of Labor Women's Bureau, 2020).[3]

The NDCP provides county-level median prices for a week of childcare by age of child, type of childcare facility (center-based or family-based child care), and year from 2008 to 2018. We draw on data from 2012-2017 to align it with the period for which childcare licensing data are available. Data for center-based care are missing for 28,542 county-year-age observations. 4,500 of the missing values correspond to observations in Indiana and New

[3]Additional states excluded from our analysis due to missing data include Rhode Island, New York, and New Hampshire. While price data are available in the NDCP for New York and New Hampshire, the specific measure we use in our analysis—median cost for childcare centers—is not provided. Rhode Island is omitted from our analysis due to an absence lack of childcare center licensing and regulation data for this state.

Mexico. Data for family-based care are missing for 30,270 county-year-age observations.

Center-based care is the most prevalent type of non-parental care used in the United States. According to the National Center for Education Statistics (NCES), the majority of children receiving non-parental care (62% based on 2019 data) are in some type of center-based care (U.S. Department of Education, National Center for Education Statistics, 2021). Due to this prevalence and limitations in the family-based child care center licensing regulations data, such as greater missingness and little to no variation where data are not missing, we consider only center-based care in this study. We assume that the costs of center-based care serve as a proxy for other types of care, supported by 90% correlation between center-based and family-based childcare costs where data are non-missing.

The NDCP warns data users that "about 80% of the childcare price data have at least one type of imputation" (Department of Labor Women's Bureau, 2020). These imputations include temporal and geographic imputations such as imputing weekly prices from annual and county-level prices from state, or assigning prices across age groupings or imputing missing data. To the extent that variations in the price data across space, time, and age groupings are due to these imputations and not genuine variations in prices, our fixed effects estimators will be spurious. To the extent that the variations in the price data across space, time, and age groupings that are correlated with variations in licensing regulations are due to these imputations and not genuine covariance, then our instrumental variables analyses will also be spurious. However, we find, as we discuss below, that our estimates are remarkably robust to research design - this robustness does not rule out spurious variation driving the results, but it does make it less likely.

Table 1 presents summary statistics of center-based childcare prices, representing the county-level median weekly costs for full-time center-based childcare for each age group in our study. Childcare costs are highest for younger age groups and decrease as children age. Weekly childcare costs for children in center-based care range from $159.0 for children below age one to $130.9 for children between ages four and five. Corresponding costs for children

in family-based childcare are $122.7 for children below age one and $112.1 for children ages four and five. Family-based care is generally less expensive than center-based care, and both types of providers show a decline in weekly costs as children age.

Figure 1 illustrates the spatial distribution of weekly center-based childcare costs by age group as a percentage of median household income across U.S. counties, highlighting the significant spatial heterogeneity in childcare expenses, with coastal regions and large U.S. metropolitan areas ranking as the most costly.

2.3 Child Care Licensing Studies

Regulations in the U.S. childcare industry are implemented to ensure that children are cared for under specific quality standards. These regulations, along with their levels of stringency, vary by state, type of provider, and child-age group. There is one set of regulations that applies to center-based care—including daycare centers, preschools, nurseries, and Head Start programs—and another set that applies to family-based daycare homes. Generally, regulations for center-based care are more stringent than those for family-based daycare homes. Since our study focuses on center-based care, we will concentrate on the regulations relevant to this category.

Within the center-based childcare category, regulations can still vary significantly, covering aspects from the required square footage per child to the educational qualifications of the staff. Following previous literature, we focus on three regulations that are most closely related to the quality of care provided: maximum child-to-staff ratios by age group, maximum class group size, and minimum education requirements for student-facing staff (Hotz and Xiao, 2011; Gorry and Thomas, 2017; Ali et al., 2024). More stringent regulations would reflect smaller child-to-staff ratios, lower maximum class group size, and the presence of staff education requirements. Although these regulations are commonly used across all U.S. states, the specific standards across states and different age groups vary.

We draw data on state-level childcare regulations from The Child Care Licensing Stud-

ies of 2011, 2012-2014, and 2017 (Fischer and Martella, 2013; Fischer and Orlowski, 2018, 2020). These studies, conducted by the National Association for Regulatory Administration (NARA) and the National Center on Early Childhood Quality Assurance, include licensing regulations on maximum child-to-staff ratios by child age and type of provider, maximum class group size limits by age of child and type of provider, and the educational requirements of child-facing staff.

While the Child Care Licensing Studies are available only at intervals (2011, 2012-2014, and 2017), they include information about when regulations changed. Using this information, we produce a largely complete panel of the licensing regulations in place between 2012 and 2017. We omit 2011 from this panel due to the absence of data on regulations prior to 2011. However, the data provided in the 2011 study are used to inform the panel for 2012 as well as subsequent years. Tables 2 and 3 provide summary statistics showing the variation of state-level licensing requirements over time and by age, respectively. Figure 3 illustrates the geographic variation of licensing requirements.

Mothers in the panel sample may have multiple children under age five, each subject to different regulatory standards based on age. To account for this, we construct a mother-level measure of regulatory stringency for the panel analysis by identifying the minimum standards applicable to each child's age and averaging them, giving equal weight to each child. This approach allows us to capture age-specific regulatory variation while aggregating it at the mother level in a meaningful way. For the cross-section analyses, we use the regulations directly in the IV approach as excluded instruments.

2.4 Other sources

Other control variables that we use in this study include the probability of getting subsidized childcare and measures of childcare market size.

To control for families that receive subsidized childcare, we construct a measure of the probability of securing a funded Head Start placement at each county-age level. Head Start

is a federal program designed to offer fully subsidized, high-quality childcare, and additional support services to children ages zero to four of families residing at or below the poverty line.

We gather state-level data on cumulative enrollment counts by children's age group from the Office of Head Start Program Information Reports (PIR), 2012-2017. PIR reports must be submitted annually by all Head Start grantees and delegated agencies. They collect detailed information on Head Start programs including total and cumulative enrollment counts, counts by child age group, and counts by type of provider.[4].

Because the lowest geographic level of variation available in our combined dataset is the county, we assign state-level funded placements to counties based on the population living at or below the poverty line at the county level. Population data come from the American Community Survey (ACS) five-year estimates covering the period of 2012 to 2017. Using this ACS data, we also obtain county-level counts of eligible children in households that qualify for Head Start, categorized by age group. By calculating the ratio of funded slots to the count of eligible children by county-age, we determine the probability of eligible children securing a funded Head Start placement in each county-age group.

While the PIR data enable us to identify funded enrollment counts by age group and type of provider (center-based vs. home or family-based), we cannot observe both categorizations simultaneously. Specifically, we cannot observe the number of funded slots in center-based care versus family-based care within age groups. As we rely on counts by age group, our data inherently include a small portion of enrollment counts that correspond to home-based care instead of center-based slots. Nevertheless, Head Start data indicate that the majority of Head Start care is provided through center-based providers.[5]

[4]Cumulative enrollment is the actual number of children that were served within a program year. That is, cumulative counts include children who left during the program year and those children who entered to replace them. Total funded enrollment counts do not account for turnover. For additional information see `https://eclkc.ohs.acf.hhs.gov/sites/default/files/pdf/no-search/service-snapshot-all-programs-2022-2023.pdf`. Accessed May 10, 2024

[5]In program year 2022-2023, 88.6% of funded enrollment corresponded to center-based care. See `https://eclkc.ohs.acf.hhs.gov/sites/default/files/pdf/no-search/service-snapshot-all-programs-2022-2023.pdf` Accessed May 10, 2024.

To control for childcare market size, we use data on the number of childcare establishments at the county level from the U.S. Census County Business Patterns (CBP) and the Non-Employer Statistics by Demographics (NES-D). The CBP provides data on employer business establishments, and the NES-D provides data on non-employer business establishments. We retain childcare-specific establishments from both sources using the childcare NAICS code 624410.

2.5 Linking data across sources

Finally, we combine the ACS-LEHD data with the childcare licensing regulations and childcare costs data. The ACS-LEHD data contain data on child's age, quarter, and county. First, we link these data to the childcare licensing regulations data (unique by child's age, quarter, and state) and then to the childcare costs data by child's age, county, and year.

To link data across all sources, we must harmonize the age of children across datasets. Each data source takes a different approach to recording the age of children, and some age categories change over time. Despite these differences, we produce largely consistent age categories across the datasets. To harmonize ages across datasets, we produce "analytical age" categories that allow us to combine data, albeit with some limitations and overlap. For example, the analytical age category of "less than 6 months" includes ACS-LEHD data covering the first two quarters after birth (months 0-6 inclusive), licensing regulations for infants six weeks to eight or ten months (depending on the year), and NDCP cost data for infants five months (inclusive) and younger. Although the categories are not perfectly aligned, the majority of the relevant months for a given age fall within the analytical age, which is why they have been assigned as shown. Figure 2 summarizes the different age categories by data source and analytical age.

Table 4 reports summary statistics for the final panel dataset, which includes 14,340,000 mother-quarter observations. On average, 60% of mothers participate in the labor force each quarter. Median quarterly childcare costs—drawn from the median county level childcare

costs available from NDCP and aggregated for each mother based on the number and age(s) of her children—average $2,180, while per-child quarterly costs average $1,703. The typical mother-quarter includes 1.8 children, 0.85 of whom are under age five. Mothers in the sample average 33 years old, and quarterly household income averages $17,140. About 75% of mothers are married, and the average probability of receiving a Head Start-funded childcare placement is 31%. The sample includes only mothers between ages 18 and 65 who had at least one child under age five at some point during the panel period.

3 Empirical Framework

Estimating the impact of childcare prices on mothers' labor force participation presents significant empirical challenges. First, unobservable characteristics at the geographic, temporal, and mother levels may simultaneously affect both childcare costs and mothers' labor force participation, potentially leading to omitted variable bias. Second, mothers' demand for non-parental paid childcare due to work obligations can also influence childcare prices, resulting in reverse causality.

To address these identification issues, we employ two complementary strategies leveraging our comprehensive dataset. First, we exploit the panel structure of our data and estimate a two-way fixed effects (TWFE) model to account for geographic, temporal, and mother heterogeneity. This approach allows us to control for national time trends and time-invariant differences in employment participation across mothers and counties, as well as additional observable characteristics. While this approach allows us to mitigate important sources of bias, we recognize that it may not fully address all potential sources of bias (i.e. selection bias).

Our second strategy to address both omitted variable bias and reverse causality is to implement an instrumental variables (IV) approach. This strategy allows us to estimate the impact of childcare costs on mothers' labor force participation using only exogenous

variation in childcare costs, allowing us to better address issues of selection bias. We present and discuss these strategies in more detail in the following sections.

3.1 Fixed Effects Model

We first estimate the effect of childcare costs on mother's labor force participation using the following fixed effects specification:

$$lfp_{icq} = \beta_0 + \beta_1 Costs_{icq} + \beta \mathbf{X_{icq}} + \alpha_y + \gamma_i + u_{icq} \tag{1}$$

where i indexes the mother, c indexes the county, q indexes the quarter, and y indexes years. lfp_{icq} is an indicator that equals one if the mother has positive earnings and zero otherwise. $Costs_{icq}$, is the childcare cost measure that we construct using two different approaches to reflect different aspects of the childcare cost burden. In one specification, $Costs_{icq}$ captures each mother's total quarterly childcare cost based on the number of her children under age five, their ages, and age-specific childcare prices in her area. This variable reflects the total financial outlay for early childcare that a mother would face in a given quarter and is a useful measure of the overall cost burden she may consider when making labor force decisions.

In a second specification, $Costs_{icq}$ captures a per-child cost estimate obtained by dividing the total childcare cost by the number of children under five in the household. This measure serves to normalize the cost across families of different sizes, making it easier to compare childcare prices across households.

Using both measures allows for a more nuanced understanding of how childcare costs affect maternal labor supply. While total cost captures the full budgetary impact, the per-child measure helps standardize the treatment and facilitates comparisons across families with differing childcare needs.

$\mathbf{X_{icq}}$ is a vector of time-varying control variables that may influence a mother's decision to enter the labor force including mother's age and its quadratic term, number of children

under five in the household, and their probability of getting a Head Start-funded seat to control for the likelihood of getting subsidized childcare.[6]

The parameter of interest is β_1, which captures the effect of childcare costs on mothers' labor force participation, yet with certain limitations that we discuss in more detail shortly. This specification exploits variation in county-age changes in childcare costs. This implies that treatment is continuous as the intensity of treatment will vary based on county trends in childcare prices and the presence and age of children. We include two-way fixed effects (TWFE) to account for spatial and time heterogeneity. α_y are year-fixed effects to control for national time trends, and γ_i are mother-fixed effects to control for time-invariant differences in labor force participation by mother, and county for the case of non-mover mothers.[7]

Due to the variation in children's birth timing and county-level price trends, our treatment not only varies in intensity but is also staggered across time. Consequently, this specification is susceptible to TWFE bias concerns highlighted in recent econometric literature (Goodman-Bacon, 2021; Borusyak et al., 2024). In brief, TWFE estimators can be biased because they incorporate comparisons between newly treated units and already treated units serving as controls. These comparisons become problematic in the presence of heterogeneous or dynamic treatment effects. The bias is exacerbated when treatment heterogeneity is most pronounced towards the middle of the panel, as this is where the TWFE estimator tends to place more weight.

Naturally, the solution to this source of bias is to prevent the estimator from making bad comparisons (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). However, settings with continuous treatment like ours are more complex to address. The economics literature on this front is rapidly evolving (Callaway et al., 2024; Borusyak et al., 2024; De Chaisemartin

[6]Our intention was to also include the share of childcare establishments (employer and non-employer) as an additional control. However, this variable was excluded due to a lack of variation at the mother level.

[7]Our data treat all mothers as non-movers since state of residence information is collected from the ACS, and backfilled to prior quarters in the panel. This will be true for the vast majority of mothers in our data, given that population mobility rates tend to be fairly small (about 15%) and declining over time, as reported by the ACS. See `https://www2.census.gov/programs-surveys/demo/visualizations/geographic-mobility/acs-1yr/residence.pdf`. Accessed March 18, 2025.

and d'Haultfoeuille, 2024), however an optimal solution for continuous treatment under our particular setting remains unclear.

It is important to note that settings with continuous treatments give rise to different parameters of interest compared to those with binary treatments. In settings with staggered binary treatment, the main focus is on "level" effects, captured by the average treatment effect of the treated (ATT) and the average treatment effect (ATE). These measure the effect of being treated versus not being treated at all (Callaway and Sant'Anna, 2021).

In contrast, continuous treatment settings necessitate the consideration of "slope" effects, which quantify the impact of marginal increases in treatment intensity (Callaway and Sant'Anna, 2021; Callaway et al., 2024). This shift introduces two new parameters of interest: the average causal response (ACR) and the average causal response on the treated (ACRT). While the ATT and ATE capture discrete changes from untreated to treated status, the ACR and ACRT measure the effect of incremental changes in treatment intensity.[8]

The ACRT and ACR provide different insights into the effects of treatment intensity changes. The ACRT focuses on units already receiving treatment, which in our context would measure how incremental changes in childcare costs affect employment outcomes for mothers currently paying for childcare. The ACR, however, offers a broader perspective by considering all units, regardless of their current treatment status.

In our study, although all mothers are treated at some point in the panel and to some degree, we could focus on recovering the ACR. This choice would allow us to capture a more general understanding of how variations in childcare costs affect maternal labor supply across our entire sample, including those with potentially very low childcare costs.

To recover the ACR, we would have to satisfy four key assumptions (Callaway et al., 2024). First, we would assume random sampling, which we believe is satisfied through our use of a nationally representative survey for data on mothers and employment-related outcomes. Second, we would assume support across the range of treatment intensities. This means that

[8]In the binary case, the effect of being treated at all is equivalent to the effect of an increase in treatment. Therefore, level and slope effects are indistinguishable in the binary treatment scenario.

for each possible level of childcare cost (our treatment), there must be units experiencing that level of cost. Our large dataset and the diverse set of mothers and household compositions represented make this assumption reasonable. Third, and perhaps most critically, we would have to assume that there are no anticipation effects. This assumption requires that mothers do not change their employment status, either on the intensive or extensive margin, in anticipation of future changes in childcare costs. We argue that this assumption holds in our context for several reasons. First, a significant portion of changes in childcare costs come from market price adjustments that individual mothers cannot anticipate. Another portion of the changes in childcare costs comes from changes in family composition. While changes due to children getting older or new births can often be anticipated, it is unlikely these anticipated changes affect a mother's work choices before the price change becomes effective. For instance: a) if a mother is currently not working due to high childcare costs, an anticipated increase in costs is unlikely to induce her to work; b) if a working mother anticipates an increase in costs, she may plan to reduce hours or leave the workforce, but is unlikely to do so before the cost increase becomes effective; c) similarly, if costs are expected to decrease, a non-working mother might plan to enter the workforce but is unlikely to do so before the cost reduction takes effect and; d) if a working mother anticipates a reduction in costs, it is unlikely for her to increase her work supply prior to the cost dropping. If there were significant anticipation effects, it would suggest that childcare costs are not as relevant to mothers' employment decisions as we hypothesize. This would likely result in statistically insignificant estimates in our analysis, effectively testing our core hypothesis.

Lastly, we would have to assume "strong parallel trends" (SPT) (Callaway et al., 2024). This assumption extends the standard parallel trends assumption to the continuous treatment setting. SPT requires that, in the absence of higher dose treatment, all units would have followed the same potential outcome path had they received any given lower dose of treatment. This allows lower-dose units to serve as valid counterfactuals for higher-dose units, enabling comparisons across different treatment intensities. While this is a strong as-

sumption, it would be necessary for identification in the continuous treatment DiD setting.

For completeness, to recover the ACRT, we would need to make an additional assumption—namely, that selection bias is zero. In our context, this selection bias arises from mothers self-selecting into different levels of treatment (i.e. childcare costs). Because parents actively select the cost of childcare they are willing to pay, the assumption of zero selection bias is difficult to justify. This in part motivates our second strategy using an instrumental variables approach.

Nevertheless, while we can still reasonably satisfy the four assumptions needed to recover the ACR, existing solutions to recover the ACR are not yet well-suited for settings like ours. We have a case of truly continuous treatment where dose groups are very small, as opposed to discrete multi-valued treatment. Callaway et al. (2024) propose a non-parametric approach to recover the parameters of interest in such settings. However, implementing a non-parametric approach with our large dataset has proven to be computationally prohibitive.

We considered two Stata packages for non-parametric estimation, lpoly and lowess, with lpoly being the one we tested more extensively. Both come with important limitations. For instance, lpoly runs kernel-weighted local polynomials for an outcome and a single regressor. This meant that we had to first residualize the model and run lpoly on the residualized predicted values. This adds complexity to an already challenging method to get the correct standard errors for inference. Moroever, to extract any estimates or related statistics from the lpoly estimation, one must define specific treatment levels at which local polynomial regressions should be estimated, undermining the goal of properly capturing the continuous nature of the treatment. Meanwhile, **lowess**, which estimates multiple weighted regressions at each point in the dataset, is computationally inefficient in our setting, as its burden scales exponentially with data size.

Given these computational challenges, we considered the possibility of binning the treatment variable and following a simpler approach where we would compare treated units of

each cost bin, with not-yet treated units. However, this approach requires assuming that, once a mother is treated at a given childcare cost level, her treatment dose remains constant over time—an assumption that is not valid in our setting.

Considering these limitations—and recognizing the potential for bias—we proceed with a TWFE approach for this first part of our analysis.

3.2 Instrumental Variables (IV) Approach

Childcare regulations are implemented by state governments to set certain standards of care and ensure quality. Each state determines which regulations to implement and their specific standards, tailoring them to their unique needs and contexts. Among the most common regulations are minimum staff-to-student ratios, maximum class-group sizes, and staff training requirements. These regulations exist to ensure that children receive sufficient attention and develop in safe, healthy environments. Given that young children require substantial personalized care, higher staff-to-student ratios and smaller class-group sizes are crucial in meeting these needs. Furthermore, extensive research underscores the importance of the first five years of life for individual development (Center on the Developing Child, 2007), highlighting the critical role that interaction with well-trained and knowledgeable early childcare professionals plays in fostering healthy development during these formative years.

Childcare regulations primarily affect organized care settings, with center-based care facing the most stringent regulations. Nonrelative family-based care providers often have fewer or less stringent regulations, while informal or relative-based care remains unregulated. The U.S. Department of Education, National Center for Education Statistics (2021) reports that 59% of children under school age not yet in kindergarten use at least one type of non-parental care arrangement, with 62% of these children enrolled in regulated childcare centers or family-based centers. This demonstrates that regulations impact a significant portion of the relevant population.

Compliance with childcare regulations incurs costs and, therefore, has an important effect on childcare prices (Blau, 2003; Heeb et al., 2004; Gorry and Thomas, 2017). Contrary to this evidence, Blau (2007) finds no effect on prices in a California study, suggesting centers may adjust along other margins when regulations aren't binding, such as reduced staff wages. This finding, however, has limited external validity and is not supported by our preliminary analysis. Studies showing a significant impact on pricing (Blau, 2003; Heeb et al., 2004; Gorry and Thomas, 2017), rely on nationally representative survey data providing more comprehensive evidence of price effects. Additionally, regulations impact childcare supply (Hotz and Xiao, 2011), which economic theory suggests would also influence prices. Brown (2018) finds that care of older children (with less stringent regulations) cross-subsidizes the care of younger children, such that when older children become eligible for public Pre-K, care slots available for younger children fall. It is on the basis of this evidence that we intend to use childcare regulations as exogenous shifters of childcare costs in an instrumental variables approach.

For childcare regulations to be valid instruments, they must satisfy two conditions: relevance and the exclusion restriction. The relevance condition requires that regulations significantly affect childcare costs, which is supported by the evidence discussed earlier. The exclusion restriction demands that regulations only affect maternal employment through their influence on prices. This condition merits closer examination.

The first potential violation relates to direct labor market effects. Changes in staff-to-student ratios or education requirements could impact labor demand and supply directly, considering that women dominate in the childcare workforce (CCAoA, 2019, p.21). To mitigate this concern, we exclude all mothers ever observed in the LEHD to have been employed in the childcare industry—to the extent possible— during our panel period from our analysis.

The second potential violation involves signaling effects of childcare regulations. More stringent regulations might signal higher quality care, potentially encouraging mothers to join

the labor force independently of price effects. This scenario presents two competing forces: increased confidence in childcare quality could boost maternal labor force participation, while the associated higher costs due to stricter regulations could decrease it. It is not clear ex-ante which of these two forces dominates.

However, the evidence suggests that the price effect dominates the signaling effect. Heeb et al. (2004) shows that when faced with more stringent regulations and consequently higher prices, mothers tend to shift away from regulated care to cheaper, less regulated options. This behavioral response indicates that the price effect outweighs any positive signaling effect on maternal employment.

This finding is crucial for two reasons. First, it supports the exclusion restriction by suggesting that the primary channel through which regulations affect maternal employment is indeed through prices. Second, it aligns with economic theory, which predicts that increased costs of childcare would, all else equal, reduce maternal labor force participation.

Furthermore, if the signaling effect were dominant, we would expect to see increased demand for regulated care following the implementation of stricter standards. Instead, the observed shift towards less regulated care options strongly suggests that the price mechanism is the primary driver of behavior changes.

In light of this evidence, we argue that while a signaling effect may exist, it is likely overshadowed by the price effect. This argument strengthens the case for the validity of our instrumental variables approach, as it supports the notion that childcare regulations primarily influence maternal employment through their impact on childcare costs.

We, therefore, exploit exogenous variation in childcare regulations to estimate the impact of childcare costs on maternal labor force participation using an instrumental variables approach (IV).

The first stage of our IV approach is specified as follows:

$$Costs_{icq} = \alpha_0 + \alpha_1 Ratio_{icq} + \alpha_2 GroupSize_{icq} + \alpha_3 EdReq_{icq} + \alpha \mathbf{X_{icq}} + \gamma_i + \phi_y + e_{icq}. \quad (2)$$

In the second stage, we estimate the impact of the costs of childcare on mothers' labor force participation as follows:

$$lfp_{icq} = \beta_0 + \beta_1 \hat{Costs}_{icq} + \beta \mathbf{X_{icq}} + \delta_i + \alpha_y + u_{icq}. \tag{3}$$

where $Costs_{icq}$ represents one of two measures of childcare costs as described in the fixed effects section. Licensing regulation data in our merged dataset is in child age-county-quarter format. To aggregate these at the mother level, we follow the approach in Blau (2003) and Hotz and Xiao (2011) and construct a regulation stringency measure for each regulation of interest at the mother level, based on her number of children and their corresponding ages. That is, for each mother in a given quarter, we average the applicable childcare regulations based on the number and ages of her children. Identification in this research design comes from variation in childcare regulations across states and quarters, interacted with differences in the age composition and number of children that mothers have within a county-quarter..

$Ratio_{icq}$ represents the child-to-staff ratio stringency measure faced by mother i in quarter q, $GroupSize_{icq}$ is the stringency measure for the maximum number of children per class faced by mother i in quarter q, and $EdReq_{icq}$ is the mother-level stringency measure for provider staff training requirements. $\mathbf{X_{icq}}$ is a vector of control variables, including mother's age, the quadratic of mother's age, the number of kids in the household under age five, and the probability of getting a Head Start-funded seat. We include mother fixed effects δ_i to control for any time-invariant unobservables at the mother level and year fixed effects α_y for economic shocks that may affect all counties and mothers equally. Standard errors are clustered at the state level.

To exploit the rich variation that the data provide, we also estimate this IV specification using a cross-section of mothers who have only one child under five years of age. We create this cross-section by retaining the mother-quarter observation where the ACS matched the LEHD, and child number and age restrictions are satisfied. This approach allows us

to exploit variation in regulations by child age group, without the need to aggregate the licensing regulation data. The identification still comes from variation in regulations at the mother-child-quarter, but at a more granular level. Furthermore, we can include additional control variables that could influence mothers' choice to join the labor force, such as mothers' educational attainment, race, ethnicity, spousal income, marital status, and the probability of getting a Head Start-funded seat to control for subsidized childcare. With this analysis, we intend to get a baseline estimate of the impact of childcare costs on mothers' labor force participation that is independent of the number of children. More details on the cross-section analysis (and the results of the cross-section analysis) are provided in the Appendix.

Throughout this analysis, we assume that the county level represents the childcare market well. Hotz and Xiao (2011) provide evidence supporting this choice, showing no significant effect in their results when exploring different market sizes, which gives us confidence in our use of the county as the childcare market.

4 Results and discussion

Table 5 reports the results from estimating the TWFE estimator in Equation 1 using each childcare cost measure described in the empirical section. As a reminder, both measures are calculated at the mother-quarter level. Columns 1 and 2 report estimates using total quarterly childcare costs, while columns 3 and 4 use per-child quarterly childcare costs.

Column 1 shows that a $100 increase in total quarterly childcare costs reduces the likelihood of a mother participating in the labor force by 0.2 percentage points, equivalent to a 0.34 percent decline relative to the sample mean of 58%. To account for potential delays in mothers' employment responses, Column 2 uses a lagged by one quarter cost measure. The estimated effect drops to a (rounded) zero, but remains negative and statistically significant at the 1% level.

The estimates based on per-child quarterly childcare costs—capturing the unit price of

care—are slightly larger in magnitude. A $100 increase in this measure reduces the likelihood of labor force participation by 0.4 percentage points, or a 0.68 percent decline from the sample mean. When using the lagged per-child cost, the estimated effect decreases to 0.1 percentage points, rather than falling to zero as in the case of the total cost measure.

A key concern in this analysis is that the continuous nature of the treatment—childcare costs—may bias estimates recovered using the TWFE approach. The direction of our estimated effects aligns with prior research, which finds that reducing childcare costs—whether through subsidies or publicly funded care—has a positive and significant effect on maternal labor force participation (Gelbach, 2002; Cascio, 2009; Gurrentz, 2022; Pepin, 2020). This consistency suggests that any bias in our estimates likely affects their magnitude rather than their sign.

Previous research shows that the largest changes in childcare costs—and corresponding shifts in maternal employment—typically occur at key transitions: when children enter the household and when all children age into the public school system (Erceg, 2023; Fitzpatrick, 2012; Herbst, 2017). These transitions, however, affect labor supply in opposite directions. By estimating effects using a panel of mothers observed across both types of transitions, the resulting impacts may partially offset each other. Moreover, because the variation in childcare costs we exploit occurs mostly between these two events, we expect our estimates to capture smaller cost changes and, therefore, to understate the full responsiveness of maternal labor force participation to childcare costs.

In addition, selection bias may further attenuate the estimated effects. Mothers who need or intend to remain in the labor force are more likely to choose a childcare arrangement that enables them to do so, particularly when facing high costs. This self-selection response mitigates the observed impact of cost variation on labor force participation, introducing an additional downward bias. Finally, we capture only the effects of the costs of childcare on the extensive margin while many adjustments are also made along the intensive margin. For this reason as well, our estimates represent a conservative estimate of the impacts of the costs

of childcare on mothers' labor force participation. Taken together, these factors reinforce the view that our findings represent a lower bound on the true effect of childcare costs on maternal labor force participation.

To further examine potential sources of bias in our estimates, we conduct a heterogeneity analysis by income level. We divide mothers into three income groups: low-income (first and second income quantiles), middle-income (third quantile), and high-income (top two quantiles). Table 6 reports TWFE estimates using per-child quarterly childcare costs as the key independent variable within each income group.

In Column 1, a $100 increase in per-child childcare costs reduces labor force participation among low-income mothers by 0.5 percentage points, representing a 1.2% decline from their average participation rate of 41%. Column 2 shows that for middle-income mothers, the same increase leads to a 0.4 percentage point decline, or a 0.63% decrease from their group mean of 64%. In Column 3, the effect for high-income mothers is smaller: a $100 increase corresponds to a 0.2 percentage point reduction in labor force participation, or a 0.27% decline from their average rate of 73%. These results indicate that low-income mothers are most responsive to increases in childcare costs, highlighting their greater vulnerability to cost fluctuations in terms of labor force participation.

To address the limitations of the TWFE estimates, we further examine the relationship between childcare costs and maternal labor force participation using exogenous variation in childcare costs, aggregated to the mother-level, through an instrumental variables (IV) approach. Table 7 reports the first-stage results, where childcare cost measures are endogenous instruments, and Table 8 presents the corresponding second-stage estimates.

We use childcare cost regulations as exogenous instruments: the child-to-staff ratio, classroom group size, and an indicator for whether childcare staff are required to meet education standards. These regulations help determine whether a childcare center qualifies for licensure and are expected to influence childcare prices. Due to limited variation in staff education requirements at the mother level, this instrument drops out of the estimation. For the re-

maining two instruments, the reported first-stage F-statistics are 32 and 36, depending on the cost measure used. These values provide evidence that the instruments are valid and relevant.

The child-to-staff ratio drives most of the exogenous variation in childcare costs. A one-unit increase in this ratio – which is to say an allowance of an additional child per adult caretaker – reduces total quarterly childcare costs by $55 and per-child quarterly costs by $50, with both effects statistically significant at the 1% level. These findings align with our hypothesis that more stringent regulations increase childcare prices by placing upward pressure on operating costs.

Second-stage results in Table 8 show that a $100 increase in either cost measure produces similar effects on maternal labor force participation. A $100 increase in total quarterly childcare costs reduces labor force participation by 1 percentage point (Column 1), equivalent to a 1.7% decline from the sample mean of 58%. Similarly, a $100 increase in per-child quarterly costs reduces participation by 1.1 percentage points (Column 2), or a 1.89% decline. The larger magnitude of these IV estimates relative to their TWFE counterparts in Table 5, Columns (1) and (3), supports our hypothesis that TWFE estimates are biased downward due to unaddressed endogeneity and attenuation from measurement error.

We also examine heterogeneous effects by income level using the IV approach. Tables 9 and 10 present results using per-child quarterly childcare costs as the endogenous cost variable. A parallel heterogeneity analysis using total quarterly childcare costs yields similar results; we report those estimates in Appendix Tables 1 and 2 .

In Column 1, a $100 increase in per-child childcare costs reduces labor force participation among low-income mothers by 1.4 percentage points, representing a 3.33% decline from their average participation rate of 42%. For middle-income mothers (Column 2), the same increase leads to a 1.0 percentage point decline, or a 1.51% decrease from their group mean of 66%. Among high-income mothers (Column 3), the effect is smaller: a $100 increase corresponds to a 0.6 percentage point reduction, or a 0.81% decline from their average participation rate

of 74%.

These findings are consistent with prior research and our TWFE estimates, confirming that low-income mothers are the most responsive to changes in childcare costs relative to higher-income groups.

A key limitation of our panel analysis is the inability to control for certain observable characteristics that likely influence mothers' labor force decisions. Some variables, such as race, remain constant over time and are absorbed by fixed effects. Others, such as marital status or spouse's income, cannot be reliably constructed into panel form with the available data. To address concerns about omitted variable bias, we test the robustness of our findings using a cross-sectional sample of single-child mothers. This restriction helps isolate the effects of childcare costs from potential confounding due to family size. While focusing on single-child households yields cleaner estimates, it also introduces a tradeoff, as labor supply responses may vary significantly by number of children.

Using this cross-sectional dataset, we estimate several versions of the IV model and find that our results remain robust when controlling for mothers' education, marital status, race and ethnicity, spouse's income, and the probability of securing a Head Start placement. The Appendix provides a detailed discussion of these cross-sectional results, and their implications.

5 Conclusion

Our study provides a timely and direct analysis of the relationship between early childcare costs and maternal labor force participation, contributing new evidence to an important and growing body of literature. Using a nationally representative panel of mothers from 2012 to 2017 and employing a range of identification strategies, we provide critical insights into how childcare expenses shape maternal labor force participation decisions.

Our first contribution lies in the direct use of variation in early childcare costs—specific

for children under age five—which enables a more precise estimation of cost effects during a period of particularly high caregiving demand. By focusing on costs associated with infant, toddler, and preschool-aged children, we complement existing research on childcare subsidies and tax credits, which are also available for families with school-aged children.

Second, by combining a comprehensive dataset on childcare market prices with nationally representative labor force data, we examine how cost burdens influence maternal labor force participation across a broader spectrum of households. This approach allows us to identify important heterogeneous effects by income level, offering a more complete picture of how childcare affordability shapes work decisions for mothers in different socioeconomic groups.

Our results align with and expand upon previous findings that link reductions in childcare costs——through subsidies, public programs, or tax credits——to increases in maternal labor force participation (Gelbach, 2002; Cascio, 2009; Baker et al., 2008). By focusing on market-based cost variation, our estimates provide a complementary perspective on the labor supply effects of early childcare costs, independent of eligibility or take-up of public assistance.

We also find that the effects of childcare costs are not uniform. Labor force responses are strongest among low-income mothers, highlighting their heightened sensitivity to cost changes and reinforcing the argument that childcare affordability presents a significant barrier to economic stability for financially vulnerable families (Havnes and Mogstad, 2011; Erceg, 2023).

In addition, our results contribute to the literature on selection bias in childcare cost studies. Prior work has noted that mothers who need or intend to stay in the workforce may select into more affordable or flexible care arrangements, potentially attenuating the observed effects of cost variation (Blau and Hagy, 1998; Tekin, 2007). Our findings support this view by demonstrating that TWFE estimates likely understate the true effect of childcare costs—an insight reinforced by the larger estimates we obtain through our instrumental variables strategy.

Beyond contributing to the academic discourse, our findings provide empirical insights

relevant to ongoing policy discussions surrounding childcare affordability, labor force participation, and equity in the workplace. In particular, our findings suggest that childcare costs represent a significant barrier to maternal labor force participation, especially for low-income families.

As debates on childcare access and affordability continue, policymakers may find these results useful in assessing the labor market effects of policies aimed at improving childcare affordability and availability. By contributing to the broader literature on labor economics and family policy, our study adds to the evidence base that informs policy decisions affecting working families and labor market participation.

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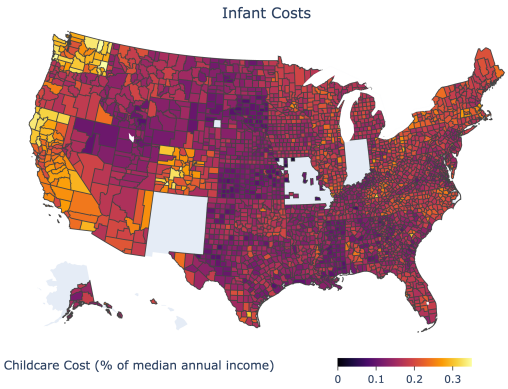
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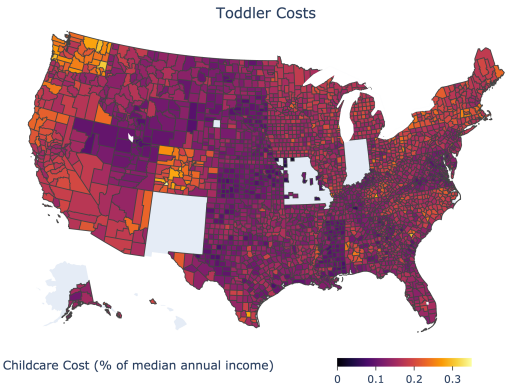
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Figure 1: **Spatial Distribution of Childcare Costs by Age**

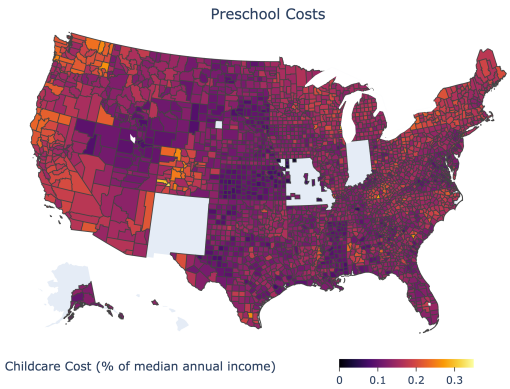
(a) Infant costs



(b) Toddler costs



(c) Preschool costs



This figure shows the spatial distribution of the average annual, center-based childcare costs across US counties as a percentage of household median income in the county. Panel (a) illustrates costs for infants (children under age 2), panel (b) illustrates costs for toddlers (children between 2 and under 3 years old), and panel (c) illustrates cost for preschoolers (children between 3 and 5 years old). Data comes from the National Database of Childcare Prices (NDCP), 2008–2018.

Figure 2: **Age Groupings by Data Source and Study Age Grouping**

NDCP	NDCP Category	CCLD 2011	CCLD 2012-14	CCLD 2017	ACS-LEHD	Analytical Approach
< =5 months	Infant	6 weeks	6 weeks	6 weeks	1-2 quarters	< 6 months
6-11 months	Infant	9 months	11 months	11 months	3-4 quarters	6 months < 1 year
12-17 months	Pre-toddler/Infant	18 months	18 months	18 months	5-6 quarters	1 to < 2 years
18-23 months					7-8 quarters	
24-29 months	Toddler	27 months	35 months	35 months	9-12 quarters	2 to < 3 years
30-35 months						
36-41 months	Preschooler	36 months	36 months	36 months	13-16 quarters	3 to <4 years
42-47 months						
48-53 months	Preschooler	48 months	48 months	48 months	17-20 quarters	4 to < 5 years
54 months to school age						

This table summarizes the different age categories by data source and the harmonized analytical age —i.e., the Analytical approach. NDCP is the National Database of Childcare Prices, CCLD is the Child Care License Data, and ACS-LEHD corresponds to our linked ACS-LEHD dataset.

Table 1: **NDCP Weekly Median Childcare Price Summary Statistics**

	Center-Based Care		Family Based Care	
	Mean	SD	Mean	SD
Childcare Cost Less than 6 months	159.0	58.02	122.7	34.51
Childcare Cost 6 months and less than 1yr	159.0	58.04	122.7	34.42
Childcare Cost 1yr and less than 2yrs	154.2	55.87	120.0	33.92
Childcare Cost 2yrs and less than 3yrs	141.3	46.27	115.5	31.25
Childcare Cost 3yrs and less than 4yrs	133.1	40.84	112.9	30.37
Childcare Cost 4yrs and less than 5yrs	130.9	39.28	112.1	30.15

Data are from the National Database of Childcare Prices (NDCP), 2012-2017. The table summarizes the average weekly median price of childcare in dollars, by analytical age group and type of care provider. Prices are adjusted to 2017 dollars using CPI adjustment factors specific to childcare.

Table 2: **Childcare Licensing Regulations by Year**

	2012		2013		2014		2015		2016		2017	
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Child-to-staff ratio	7.4	-3.71	7.4	-3.71	7.5	-3.7	7.5	-3.7	7.4	-3.7	7.4	-3.69
Group Size	10.9	-8.44	10.9	-8.44	11	-8.49	11.2	-8.46	11.4	-8.42	11.4	-8.42
Staff Training Req.	0.7	-0.48	0.7	-0.47	0.7	-0.47	0.7	-0.47	0.7	-0.47	0.7	-0.47

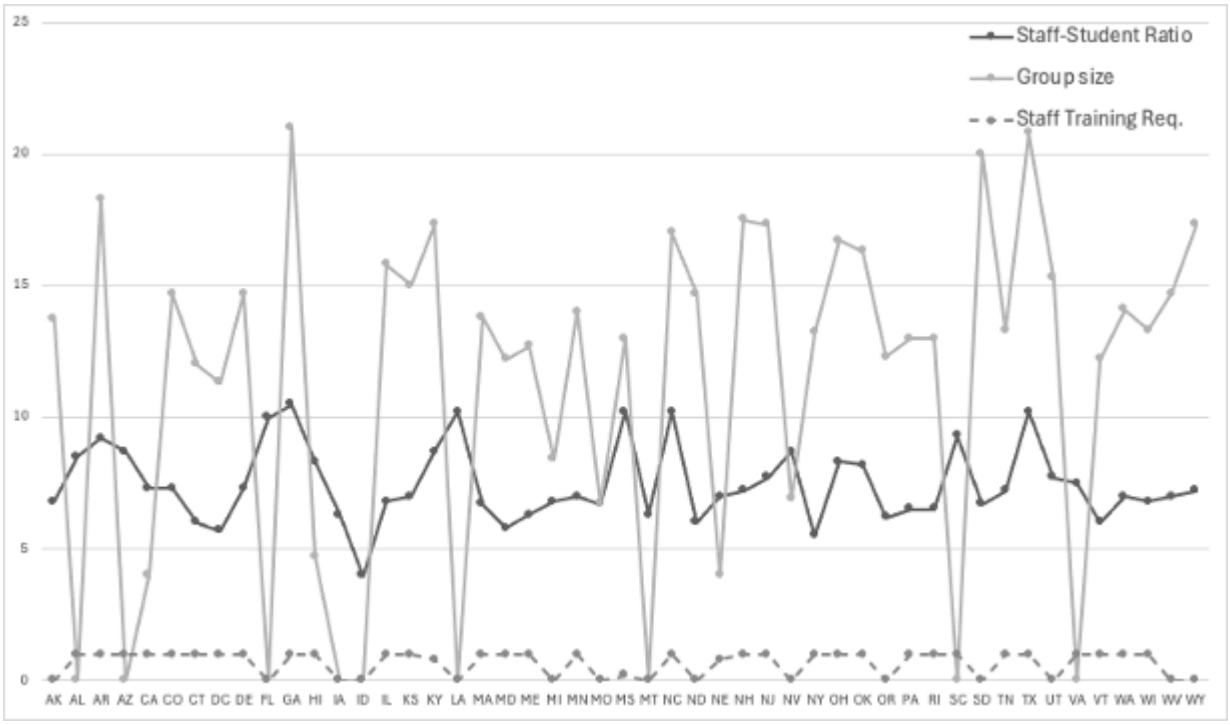
This table presents average state childcare regulations by year from 2012 to 2017. The Child-to-staff ratio is the maximum number of children per caregiver. The Group size is the maximum number of children allowed in a group. Staff training requirement is the share of states that have minimum education requirements in place. All reported values correspond to yearly means. Data are from the Child Care Licensing Studies of 2011, 2012-2014, and 2017 (Fischer and Martella, 2013; Fischer and Orlowski, 2018, 2020)

Table 3: **Childcare Licensing Regulations by Age**

	<6m		6m and <1y		1y and <2y		2y and <3y		3y and <4y		4y and <5y	
	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd
Child-to-staff ratio	4.3	0.64	4.3	0.68	5.5	1.43	7.5	2.07	10.6	2.25	12.7	3.2
Group Size	7.4	4.40	7.4	4.42	9.4	5.68	11.4	7.09	14.9	10.08	16.6	11.44
Staff Training Req.	0.7	0.47	0.7	0.47	0.7	0.47	0.7	0.47	0.7	0.47	0.7	0.47

This table presents the variation in state childcare regulations by analytical age group. The Child-to-staff ratio is the maximum number of children per caregiver. The Group size is the maximum number of children allowed in a group. Staff training is the share of states that have minimum education requirements in place. Data are from the Child Care Licensing Studies of 2011, 2012-2014, and 2017 (Fischer and Martella, 2013; Fischer and Orlowski, 2018, 2020)

Figure 3: **Childcare Licensing Regulations by State**



This plot shows the variation in childcare licensing regulations by state. The Child-to-staff ratio is the maximum number of children per caregiver by state and age of child. The Group size is the maximum number of children allowed in a group by state and age of child. Staff training is the share of states that have minimum education requirements in place. Data are state aggregates from 2012 to 2017. Data are from the Child Care Licensing Studies of 2011, 2012-2014, and 2017 (Fischer and Martella, 2013; Fischer and Orlowski, 2018, 2020)

Table 4: **2012-2017 Panel summary statistics**

	Mean	SD
Mother in the labor force	.605	.488
Quarterly childcare cost	21.800	21.180
Quarterly childcare cost per child	17.030	14.160
Total number of kids	1.839	1.147
Kids less than 5	.850	.734
Kids 5 or greater	.989	1.097
Mother's age	33.150	6.515
Mother and spouse income	17140	20020
Married	.746	.435
Probability of Head Start seat	.314	.339
Observations	14340000	

This table presents summary statistics for the 2012–2017 panel constructed by linking ACS-LEHD data with childcare licensing regulations and the National Database of Childcare Prices. The panel includes mothers who had at least one child under age five at some point during the panel period. Mothers who worked in the childcare industry during 2012-2017 were dropped from the sample. It is an unbalanced panel in which all mothers enter in the first quarter of 2012 and exit at different points based on their ACS survey date. The sample excludes observations from five states—Indiana, New Mexico, New York, Rhode Island, and New Hampshire—due to missing childcare price data. Childcare cost variables are scaled by 100 and adjusted to 2017 dollars using CPI adjustment factors specific to childcare. Observations correspond to mother-quarters.

Table 5: **Childcare Costs and Mothers' Labor Force Participation - TWFE Estimates**

	(1)	(2)	(3)	(4)
Quarterly childcare cost	-0.002***			
	(0.000)			
L.Quarterly childcare cost		-0.000***		
		(0.000)		
Quarterly childcare cost/child			-0.004***	
			(0.000)	
L.Quarterly childcare cost/child				-0.001***
				(0.000)
Mother's age	0.063***	0.064***	0.063***	0.064***
	(0.002)	(0.002)	(0.002)	(0.002)
Mother's age^2	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Kids less than 5	-0.001	-0.031***	-0.041***	-0.040***
	(0.002)	(0.001)	(0.001)	(0.001)
Probability of Head Start seat	0.018***	0.018***	0.017***	0.018***
	(0.001)	(0.001)	(0.001)	(0.001)
Mother, Year FEs	Yes	Yes	Yes	Yes
Observations	9485000	8701000	9485000	8701000
Mean Y	.58	.58	.58	.58

This table reports TWFE estimates of mothers' labor force participation. Cost variables are scaled by 100 and adjusted to 2017 dollars using childcare-specific CPI adjustment factors. *L.Quarterly childcare cost* and *L.Quarterly childcare cost/child* refer to the one-quarter lagged values of total and per-child quarterly childcare costs, respectively. *Mean Y* indicates the average labor force participation rate of mothers in the sample. Regressions are weighted using ACS person weights. Standard errors are clustered at the county level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 6: **Childcare Costs and Mothers' Labor Force Participation by Income Group- TWFE Estimates**

	Low-Income (1)	Mid-Income (2)	High-Income (3)
Quarterly childcare cost/child	-0.005***	-0.004***	-0.002***
	(0.000)	(0.000)	(0.000)
Mother's age	0.075***	-0.037***	-0.020***
	(0.003)	(0.003)	(0.002)
Mother's age^2	-0.001***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)
Kids less than 5	-0.039***	-0.037***	-0.030***
	(0.001)	(0.001)	(0.001)
Probability of Head Start seat	0.012***	0.016***	0.016***
	(0.001)	(0.001)	(0.001)
Mother, Year FEs	Yes	Yes	Yes
Observations	3642000	1836000	3831000
Mean Y	.41	.64	.73

This table reports TWFE estimates of mothers' labor force participation by income group. Column (1) presents estimates for low-income mothers (bottom two income quantiles), Column (2) for middle-income mothers (third quantile), and Column (3) for high-income mothers (top two quantiles). *Mean Y* indicates the average labor force participation rate for each group. Childcare costs are scaled by 100 and adjusted for inflation to 2017 dollars using childcare-specific CPI adjustment factors. Regressions are weighted using ACS person weights. Standard errors are clustered at the county level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 7: **IV First-Stage Results**

	Qtrly Childcare Cost (1)	Qtrly Childcare Cost/Child (2)
Mother child-to-staff ratio	-0.550***	-0.501***
	(0.072)	(0.064)
Mother average group size	-0.019	0.005
	(0.041)	(0.036)
F statistic	36	32
Observations	9485000	9485000

This table reports first-stage results from the baseline instrumental variable (IV) regressions. *Mother child-to-staff ratio* represents the average applicable child-to-staff ratio for each mother in a given quarter, based on her state of residence, number of children under age five, and their ages. *Mother average group size* represents the corresponding class group size under the same criteria. The reported F-statistics refer to the excluded instruments shown in the table. Additional controls consistent with those used in the second stage were included in the estimation but are omitted here for brevity. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 8: **Childcare Costs and Mothers' Labor Force Participation - IV Estimates**

	(1)	(2)
Quarterly childcare cost	-0.010***	
	(0.001)	
Quarterly childcare cost/child		-0.011***
		(0.001)
Mother's age	0.055***	0.054***
	(0.003)	(0.003)
Mother's age^2	-0.001***	-0.001***
	(0.000)	(0.000)
Kids less than 5	0.211***	-0.041***
	(0.023)	(0.001)
Probability of Head Start seat	0.013***	0.013***
	(0.002)	(0.002)
Mother, Year FEs	Yes	Yes
Observations	9485000	9485000
Mean Y	.58	.58

This table presents baseline instrumental variable (IV) estimates of mothers' labor force participation. Childcare costs are scaled by 100 and adjusted for inflation to 2017 dollars using childcare-specific CPI adjustment factors. *Mean Y* indicates the average labor force participation rate of mothers in the panel. Observations correspond to mother-quarters. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 9: **IV First-Stage Results by Income Level**

	Quarterly Childcare Cost/Child		
	Low-Income	Mid-Income	High-Income
Mother child-to-staff ratio	-0.450***	-0.429***	-0.462***
	(0.018)	(0.015)	(0.019)
Mother class group size	0.007	0.000	-0.004
	(0.010)	(0.009)	(0.012)
F Statistic	304	409	321
Observations	2313000	1180000	2551000

This table reports first-stage results for instrumental variable (IV) estimates by income group, where the cost measure is the per-child quarterly childcare cost. *Mother child-to-staff ratio* represents the average applicable child-to-staff ratio for each mother in a given quarter, based on her state of residence, number of children under age five, and their ages. *Mother average group size* represents the corresponding class group size under the same criteria. The reported F-statistics refer to the excluded instruments shown in the table. Additional controls consistent with those used in the second stage were included in the estimation but are omitted here for brevity. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 10: **Per-Child Childcare Costs and Mothers' Labor Force Participation by Income Group - IV Estimates**

	Low-Income	Mid-Income	High-Income
Quarterly childcare cost/child	-0.014***	-0.010***	-0.006***
	(0.001)	(0.001)	(0.000)
Mother's age	0.072***	-0.025***	-0.010***
	(0.003)	(0.003)	(0.002)
Mother's age^2	-0.001***	0.000	-0.000*
	(0.000)	(0.000)	(0.000)
Kids less than 5	-0.041***	-0.036***	-0.030***
	(0.001)	(0.001)	(0.001)
Probability of Head Start seat	0.009***	0.015***	0.016***
	(0.002)	(0.002)	(0.001)
Mother, Year FEs	Yes	Yes	Yes
Observations	2313000	1180000	2551000
Mean Y	.42	.66	.74

This table reports baseline instrumental variable (IV) estimates of mothers' labor force participation by income group. Childcare costs are scaled by 100 and adjusted to 2017 dollars using childcare-specific CPI adjustment factors. *Mean Y* indicates the average labor force participation rate for each income group. Observations correspond to mother-quarters. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

A Appendix

A.1 IV Income Heterogeneity Analysis Using Quarterly Childcare Costs

Table 1: **IV First-Stage Results by Income Level**

	Quarterly Childcare Cost		
	Low-Income	Mid-Income	High-Income
Mother child-to-staff ratio	-0.511***	-0.491***	-0.535***
	(0.021)	(0.017)	(0.024)
Mother class group size	-0.017	-0.027***	-0.037***
	(0.011)	(0.010)	(0.014)
F Statistic	296	397	264
Observations	2313000	1180000	2551000

This table reports first-stage results for instrumental variable (IV) estimates by income group, where the cost measure is the quarterly childcare cost. *Mother child-to-staff ratio* represents the average applicable child-to-staff ratio for each mother in a given quarter, based on her state of residence, number of children under age five, and their ages. *Mother group size* represents the corresponding class group size under the same criteria. The reported F-statistics refer to the excluded instruments shown in the table. Additional controls consistent with those used in the second stage were included in the estimation but are omitted here for brevity. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 2: **Childcare Quarterly Costs and Mothers' Labor Force Participation by Income Group - IV Estimates**

	Low-Income	Mid-Income	High-Income
Quarterly childcare cost	-0.012***	-0.008***	-0.005***
	(0.001)	(0.001)	(0.000)
Mother's age	0.072***	-0.025***	-0.010***
	(0.003)	(0.003)	(0.003)
Mother's age^2	-0.001***	0.000**	-0.000
	(0.000)	(0.000)	(0.000)
Kids less than 5	0.258***	0.167***	0.097***
	(0.015)	(0.013)	(0.008)
Probability of Head Start seat	0.009***	0.013***	0.014***
	(0.002)	(0.002)	(0.001)
Mother, Year FEs	Yes	Yes	Yes
Observations	2313000	1180000	2551000
Mean Y	.42	.66	.74

This table reports baseline instrumental variable (IV) estimates of mothers' labor force participation by income group. Childcare costs are scaled by 100 and adjusted to 2017 dollars using childcare-specific CPI adjustment factors. *Mean Y* indicates the average labor force participation rate for each income group. Observations correspond to mother-quarters. Regressions are weighted using ACS person weights. Standard errors are clustered at the state level and reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

A.2 Robustness Check with Cross-Sectional Data

As a robustness check, we estimate the impact of the costs of childcare on mothers' labor force participation in the cross-section generated by linking the ACS with the LEHD at the time of the ACS interview. We estimate both a simple (endogenous) linear probability model, as well as an instrumental variable regression in this cross-section. Estimating the relationship between childcare and labor force participation in the cross-section allows us to control for, and estimate the impact of, a number of mother-specific and time varying characteristics we cannot observe in the fixed effects approach, including race/ethnicity, education, marital status, and spousal income.

We restrict our cross section analysis to mothers who have one child, and only one child, under the age of five at the time of the ACS interview. This restriction means that there are fewer sources of variation in the cost of childcare in this robustness check when compared with the main analysis. Instead of allowing the costs of childcare within a household to vary with a potentially increasing number of children and their changing ages, in this analysis, household level costs vary only by the static age of the child at the time of the survey. Restricting the analysis to a single child also means that we do not need to aggregate our exogenous instruments within a household but can instead directly apply the age-appropriate child-to-staff ratio, group size, and educational licensing requirements for the child's age as exogenous instruments.

Similar to the main analysis, we impose the restriction that the mothers included in this analysis have a child at the time of the ACS-LEHD linkage. However, this restriction carries an additional limitation in the cross-section approach: we cannot estimate the impact on labor force participation of a household moving from zero children to one child. That impact can only be estimated in the fixed effects model. Also as in the panel analysis, we remove from the analytical sample mothers who were observed in the LEHD to have worked in the childcare industry during the time frame of our analysis (2012-2017).

A key difference between research design and data inputs in the cross-section analysis as

compared with the panel analysis is the source of the poverty status/income data for analysis of treatment heterogeneity across income groups and the use of the data on the probability of obtaining a Headstart seat. While in the panel analysis we rely on data from the LEHD to partition households into low, middle, and high income groups, in the cross-section we rely on the ACS designation of the household falling below or above the federal poverty line based on their income and household size at the time of the ACS survey. Because access to Headstart, and associated work requirements for Headstart participation, is only available to those below the federal poverty line, we are able to more carefully use these data in the cross-section analysis as we detail below.

To examine the cross-section estimate of the relationship between mothers' labor force participation and childcare costs, we begin by estimating a naive linear probability model, in which we fail to address the endogeneity of childcare costs and mothers' labor force decision, of the following form:

$$lfp_{icq} = \beta_0 + \beta_1 Costs_{icq} + \beta \mathbf{X_{icq}} + \delta_s + \alpha_y + u_{icq}. \tag{4}$$

where $Costs_{icq}$ represents the childcare costs faced by mother, i, in quarter, q, and county, c, and lfp_{icq} captures her labor force participation outcome (either 0, if she has negative earnings or 1, if she has positive earnings) in that county-quarter. $\mathbf{X_{icq}}$ is a set of control variables including the age and squared age of the mother, the mother's race/ethnicity, education, marital status, and her spouse's income. We estimate the naive linear probability model for those households below and above the federal poverty line separately. For those households below the poverty line, the variable, "Probability of obtaining a Headstart seat" is an additional control variable as, due to Headstart access work requirements, it may be correlated with mothers' decision to work. For those households above the poverty line, "Probability of obtaining a Headstart seat" is only relevant in so far as it exogenously shifts the costs of childcare in a given county-quarter; therefore, the variable is used as an exogenous

instrument in the instrumental variables regressions that follow for the households above the poverty line but is omitted from the naive regression here.

To estimate the causal impact of the costs of childcare on mothers' labor force participation, we rely on exogenous shifters of childcare costs that are drawn from changes in state-year childcare center licensing regulations - including child-to-staff ratios, group size maximums, and presence of educational requirements for teaching staff - and tabulations of the number of employer and non-employer childcare establishments in a given county-year. As noted above, we also include "Probability of obtaining a Headstart seat" as an exogenous instrument in the subsample of the cross-section data that is above the federal poverty line.

We estimate the causal relationship between the costs of childcare and mothers' labor force participation via an instrumental variables approach. The first stage of our IV approach is defined as follows for the above poverty subsample of the cross section data:

$$Costs_{icq} = \alpha_0 + \alpha_1 Ratio_{isq} + \alpha_2 GroupSize_{isq} + \alpha_3 EdReq_{isq} + \alpha_4 NonEmpEst_{cy} + \alpha_5 EmpEst_{cy} + \alpha_6 ProbHeadStart_{icq} + \alpha \mathbf{X_{icq}} + \gamma_s + \phi_y + e_{icq}. \quad (5)$$

where $Costs_{icq}$ for mother, i, in county, c, and in quarter, q, are estimated as a function of licensing regulations that vary by age of child, quarter, and state, the number of employer and non-employer childcare establishments that vary by county and year, and the probability of obtaining a funded Headstart seat, which varies by county and year. $\mathbf{X_{icq}}$ is again the set of control variables including the age and squared age of the mother, her race/ethnicity, education, marital status, and spouse's income. In the case of the below poverty subsample, $ProbHeadStart_{cq}$ is removed from the set of instruments and instead joins the set of control variables.

The second stage is again estimated as a linear probability model with state and year

fixed effects:

$$lfp_{icq} = \beta_0 + \beta_1 \hat{Costs}_{icq} + \beta \mathbf{X_{icq}} + \delta_s + \alpha_y + u_{icq}. \quad (6)$$

Summary statistics for the cross-section sample are shown in Appendix Table 3. From these summary statistics we can see that the proportion of mothers who are in the labor force is greater in the cross section (66 percent) than it is in the panel sample (61 percent). We can also see that the per child quarterly costs of childcare are greater in the cross section ($26.84) than the panel ($17.03). In addition, the average mother is younger and slightly less likely to be married in the cross section than in the panel.

We also report summary statistics for the exogenous instruments including child-to-staff ratios (mean of 8.1 children to each adult with a standard deviation of 3.9), class group size maximum (mean of 11 with a standard deviation of 8.9), educational requirements (present in 80 percent of state-years with a standard deviation of 0.40), and the number of non-employer (2997 (6385)) and employer (265.9 (385)) establishments per county. Finally, in each of the cross section regressions, we partition the data based on whether the household falls above or below the federal poverty line. Of the mothers observed in the cross section sample, 17.4 percent fall below the federal poverty line.

Table 3: **Cross-Section Summary Statistics, 2012-2017**

	Mean	SD
In the labor force	.6591	.474
Quarterly childcare cost	26.84	9.643
Mother's age	30.57	6.301
Black Non-Hispanic	.1096	.3124
Hispanic	.1556	.3625
High school graduate	.5055	.5
College	.4419	.4966
Married	.6943	.4607
Spouse income	5.052	4.794
Probability of Head Start seat	.2911	.3184
Child-to-staff ratio	8.084	3.907
Class group size	11.28	8.871
Staff training requirement	.7973	.402
Non-employer establishments	2997	6385
Employer establishments	265.9	385
In poverty	.1739	.379
Observations	223000	

This table displays summary statistics for the cross-section sample generated by linking ACS mothers with one child under the age of five to the LEHD. Mothers who worked in the childcare industry during 2012-2017 were dropped from the sample. The variables displayed here include the dependent variable, "In the labor force", the independent variable of interest, "Quarterly childcare costs", the control variables ("Mother's age" through "Spouse income") exogenous instruments ("Child-to-staff ratio" through "Employer establishments") the "In poverty" indicator along which the analytical sample is partitioned, and a variable, "Probability of Head Start seat" that serves as a control for the "Below Poverty" partition and as an exogenous instrument for the "Above Poverty" partition.

Results for the first cross-section specification, the naive linear probability model, are provided in Table 4. For both the above and below poverty subsamples, the impact of a $100 increase in the quarterly costs of childcare corresponds with a 0.3 pp decrease in mothers' labor force participation, representing a 0.63 percent decrease in mean labor force participation for the below poverty subgroup and a 0.43 percent decrease in mean labor force participation for the above poverty subgroup. As discussed in the body of the paper, the naive estimates presented here are downward biased due to the reverse causality of mothers' labor force participation on childcare prices.

To address this source of endogeneity, we estimate the impact of childcare costs on mothers' labor force participation using instrumental variables analysis. The first stage estimates of the IV approach are presented in Table 5 where we see a strong correlation between the exogenous instruments and the endogenous variable of interest, quarterly childcare costs. The analysis is again partitioned by poverty status and the probability of obtaining a head start seat is included as an exogenous instrument in the first stage for the below poverty group (note that this coefficient estimate is missing from column (1) due to its accidental omission from the Census disclosure review process; this will be resolved as soon as possible). F-statistics are reported for the exogenous instruments; their values indicate a very strong set of instruments.

Finally, the second stage of the IV approach is reported in Table 6 where columns (1) and (2) present the results with robust standard errors and columns (3) and (4) present the results with state-cluster robust standard errors. We present both the robust and cluster-robust SEs in this table because, while state level cluster robust SEs are appropriate for this analysis, some cluster cell counts were too small to generate reliable cluster-robust SEs. Therefore, we present columns (3) and (4) only to demonstrate that the results remain statistically significant even with cluster robust SEs. The discussion that follows will focus on columns (1) and (2).

Consistent with the panel based estimates, the cross-section estimates are statistically

significant and negative. In column (1), the below poverty subsample, a $100 increase in quarterly childcare costs induces a 1.3pp decrease in mothers' labor force participation - a result very similar to the 1.4pp decrease in the low-income group in the panel IV analysis reported in Table 10. In column (2), the above poverty subsample, a $100 increase in quarterly childcare costs induces a 0.5pp decrease in mothers' labor force participation - a result very similar to the 0.6pp decrease in the high-income group in the panel IV analysis reported in Table 10.

Overall the cross-section IV results support the findings in our preferred specification, the panel IV analysis: the impacts of the cost of childcare are statistically significant, negative, and greater for lower than for higher income households.

Table 4: **Childcare Costs and Mothers' Labor Force Participation - Naive Linear Probability Model with Cross-Section of Single-Child Mothers**

	Below Poverty (1)	Above Poverty (2)
Quarterly childcare costs	-0.003***	-0.003***
	(0.001)	(0.000)
Mother's age	-0.011***	0.036***
	(0.003)	(0.001)
Mother's age^2	0.000	-0.001***
	(0.000)	(0.000)
Black Non-Hispanic	0.130***	0.050***
	(0.009)	(0.004)
Hispanic	0.050***	0.015***
	(0.009)	(0.004)
High school Graduate	0.128***	0.184***
	(0.009)	(0.007)
College	0.136***	0.265***
	(0.014)	(0.007)
Married	-0.135***	-0.260***
	(0.011)	(0.003)
Spouse income	0.001	0.007***
	(0.002)	(0.000)
Probability of Head Start seat	0.014	
	(0.011)	
State, Year FEs	Yes	Yes
Robust SEs	Yes	Yes
Observations	35500	188000
Mean Y	.48	.7

This table presents the naive estimates of the impact of childcare costs on the likelihood of mother's labor force participation, using the cross-section of single-child ACS mothers, 2012-2017. Column (1) corresponds to the subsample of mothers who are at or below the poverty line, and column (2) presents estimates for the subsample of mothers who are above the poverty line. Childcare costs are in hundreds and adjusted for inflation to 2017 dollars using the childcare CPI adjustment factors. Mean Y represents the mean labor force participation rate for mothers in the sample. Observations correspond to mothers in a specific quarter. Regressions are weighted using ACS person weights. Standard errors are robust standard errors and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 5: **IV First-Stage Results**

	Below Poverty (1)	Above Poverty (2)
Child-to-staff ratio	-0.552***	-0.668***
	(0.010)	(0.004)
Class group size	0.018***	0.019***
	(0.006)	(0.003)
Staff training requirement	0.088	0.196**
	(0.202)	(0.099)
Non-employer establishments	-0.001***	-0.001***
	(0.000)	(0.000)
Employer establishments	0.015***	0.017***
	(0.000)	(0.000)
Probability of Head Start seat*		
Observations	35500	188000
F Stat	1362	11450

This table presents first-stage estimates of exogenous regressors on childcare costs. Child-to-Staff ratio represents the ratio of providers to children. Class group size corresponds to the number of children permitted in a classroom. Staff training requirement is an indicator of whether childcare staff are required to have specific education credentials in order for the childcare center to be licensed. Non-employer establishments are non-employer childcare establishments in a given state. Probability of Head Start seat is the probability of a mother getting a Head Start funded childcare spot. The reported F statistic corresponds only to exogenous instruments -i.e. variables in this table. Additional controls are included in this first-stage estimation including mother's age, the quadratic of mother's age, controls for race, education, marital status and spousal income, but are excluded from this table for simplicity. Column (1) presents first-stage coefficients for the subsample of mothers that are at or below the poverty line, and column (2) presents corresponding estimates for the subsample of mothers who are above the poverty line. Observations correspond to mothers in a specific quarter. Regressions are weighted using ACS person weights. Standard errors are robust standard errors and reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.

Table 6: **Childcare Costs and Mothers' Labor Force Participation - IV with Cross-Section of Single-Child Mothers**

	Below Poverty (1)	Above Poverty (2)	Below Poverty (3)	Above Poverty (4)
Quarterly childcare cost	-0.013***	-0.005***	-0.013***	-0.005***
	(0.001)	(0.000)	(0.003)	(0.001)
Mother's age	-0.010***	0.037***	-0.010***	0.037***
	(0.003)	(0.001)	(0.004)	(0.002)
Mother's age^2	0.000	-0.001***	0.000	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Black Non-Hispanic	0.145***	0.051***	0.145***	0.051***
	(0.009)	(0.004)	(0.016)	(0.008)
Hispanic	0.059***	0.015***	0.059***	0.015
	(0.010)	(0.004)	(0.010)	(0.012)
High school graduate	0.127***	0.183***	0.127***	0.183***
	(0.009)	(0.007)	(0.010)	(0.014)
College	0.154***	0.270***	0.154***	0.270***
	(0.014)	(0.007)	(0.014)	(0.013)
Married	-0.131***	-0.258***	-0.131***	-0.258***
	(0.011)	(0.003)	(0.014)	(0.009)
Spouse income	0.001	0.008***	0.001	0.008***
	(0.002)	(0.000)	(0.002)	(0.001)
Probability of Head Start seat	-0.017		-0.017	
	(0.011)		(0.012)	
State, Year FEs	Yes	Yes	Yes	Yes
Robust SEs	Yes	Yes	No	No
Cluster Robust SEs	No	No	Yes	Yes
Observations	35500	188000	35500	188000
Mean Y	.48	.7	.48	.7

This table presents the IV estimates of increases in childcare costs on the likelihood of mothers' labor force participation, using the cross-section of single-child ACS mothers, 2012-2017. Columns (1) and (3) present estimates for the subsample of mothers who are at or below the poverty line, and Columns (2) and (4) present estimates for the subsample of mothers who are above the poverty line. Childcare costs are in hundreds and adjusted for inflation to 2017 dollars using the childcare CPI adjustment factors. Mean Y represents the mean labor force participation rate for mothers in the sample. Observations correspond to mothers in a specific quarter. Regressions are weighted using ACS person weights. Models (1) and (2) use robust standard errors, and models (3) and (4) use clustered standard errors at the state level. Standard errors are reported in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, ***, respectively.